

Task 1 Motor Drive

1. Introduction:

An H-Bridge motor driver circuit is an essential building block for controlling DC motors in robotics, automation, and embedded systems. It allows the motor to rotate in both clockwise and counter-clockwise directions by controlling the polarity of the voltage applied across the motor terminals.

2. Circuit Description:

The circuit I used consists of:

- Four NPN transistors
- 9V DC supply
- DC motor
- Resistors
- SPDT switch
- Potentiometer

3.Component Selection:

- **NPN Transistors:**

- Chosen for their wide availability, low cost, and ability to handle moderate current
- Suitable for switching applications such as driving small DC motors.

- **Resistors:**

- Provide current limiting to protect the transistor bases.
- Ensure proper transistor switching without overloading the control signal.

- **DC Motor:**

- Load to demonstrate the H-Bridge's capability.
- Represents a typical small robotics motor that requires bidirectional control.

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- **9V Battery:**

- Simple and portable power source.
- Provides enough voltage to drive both the transistors and the DC motor effectively.

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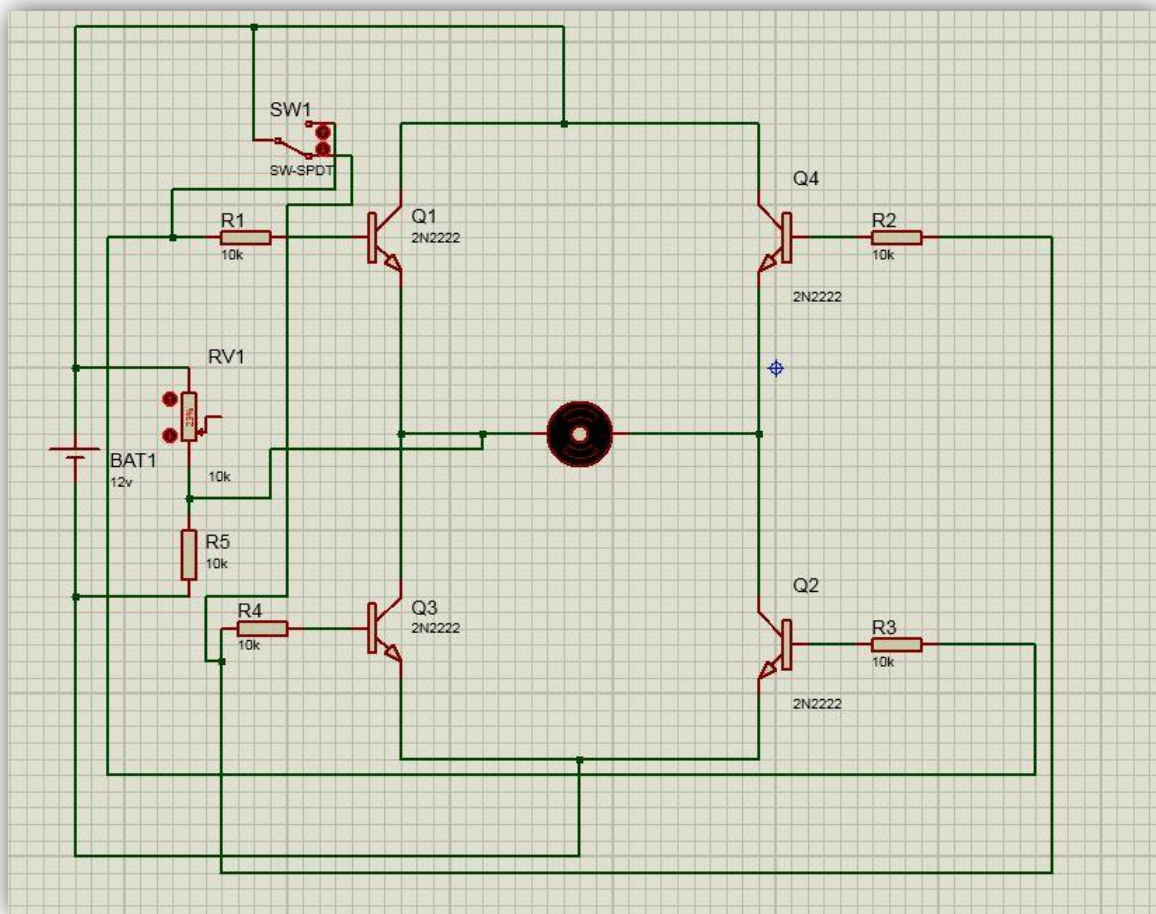
- **SPDT Switch:**

- Manually selects the motor direction by enabling different transistor pairs.

- **Potentiometer:**

- To control the motor's speed

5.The Circuit:



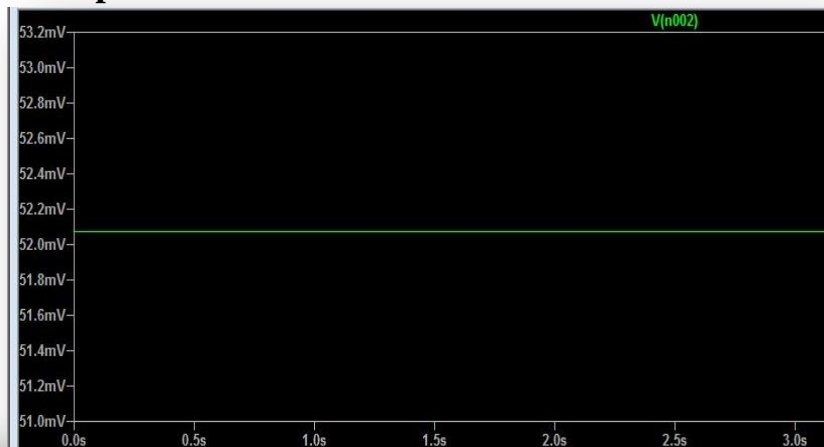
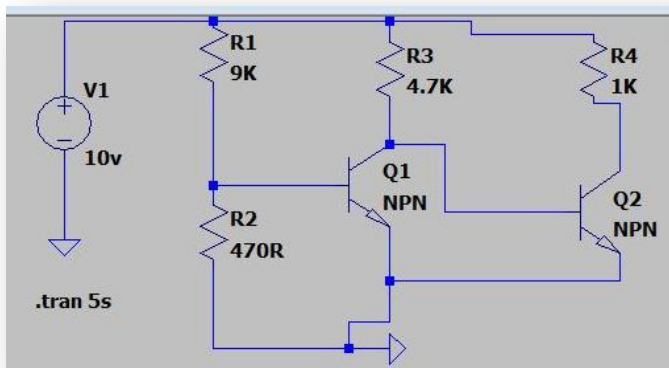
Task 2 Overvoltage Protection Circuits

1. Introduction

There are many types of overvoltage protection circuits. We are going to talk about only 2: NPN and PNP overvoltage protection circuits. We used NPN transistors to clamp a voltage. On the other hand, we used a PNP transistor in a regulator configuration to provide a stable output.

2. Circuit Descriptions

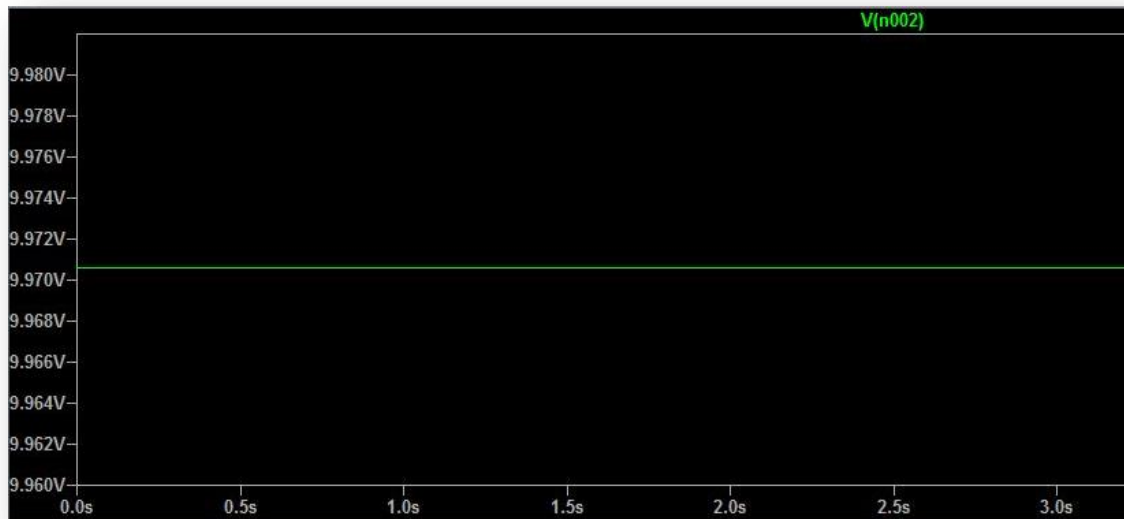
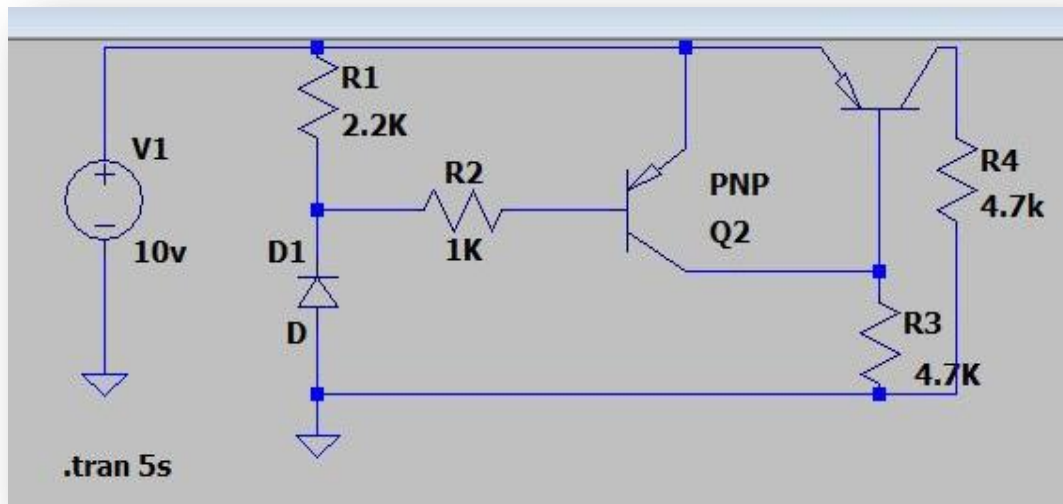
2.1 Circuit 1: NPN-Based Voltage Clamp



- **Key Components:** The main components are two NPN transistors along with a voltage divider formed by a 9k ohm and a 4.7k ohm resistor.
- **How It Works:** In normal operation, the output voltage is low enough that transistor Q2 remains off, allowing Q1 to pass the input voltage through. However, if the output voltage rises too high, the voltage divider turns Q2 on. This action robs Q1 of its driving current, forcing it to reduce conduction and effectively clamp the output voltage.

2.2 Circuit 2: PNP-Based Voltage Regulator

- **Key Components:** The design centers around a PNP transistor (Q2) as the pass element, a Zener diode (D1) for a voltage reference, and a feedback network of two $4.7\text{k}\Omega$ resistors.
- **How It Works:** The Zener diode provides a fixed reference voltage. The resistor network samples the output voltage and feeds it back to the base of the PNP transistor. This creates a closed feedback loop that continuously adjusts the transistor's conduction to counteract any changes in the output, holding it at a very stable output voltage.



Conclusion:

In a nutshell, these are two different tools for two different jobs. The NPN circuit is a **voltage clamp**, a protective device that acts as a safeguard against faults. The PNP circuit is a **voltage regulator**, a performance-oriented device designed for continuous, stable operation.